The impact of the ELIXIR community in Machine Learning

ELIXIR All Hands , 10-12 June 2024, Uppsala, Sweden

Overview of the Machine Learning Focus Group

With the continuous generation, processing, and transformation of biological data, the application of Machine Learning (ML) has become one of the most **effective approaches for extracting insights** from this data and **supporting decision-making processes**.

The ELIXIR Machine Learning Focus Group launched in **October 2019**, specifically to capture the emerging needs in ML expertise across the community, and has been consistently producing high-impact outputs over the past 5 years.

1. DOME Recommendations Task

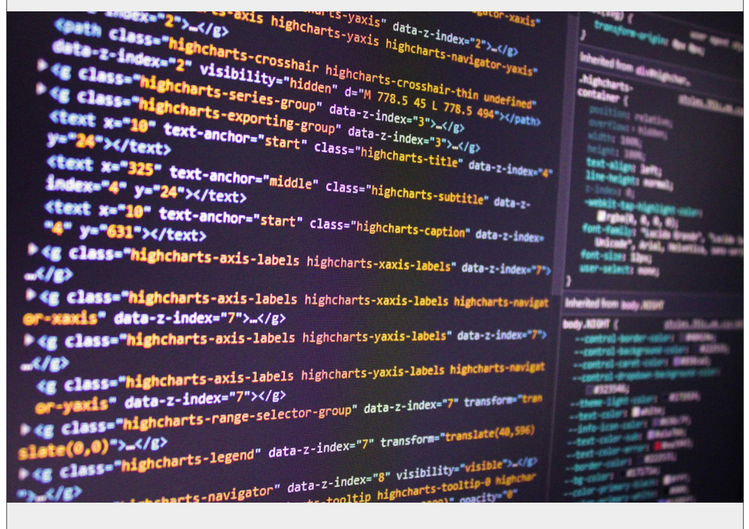
The **first major output** of the Focus Group is the publication of the DOME recommendations.

These are a set of **community-wide recommendations** for reporting supervised ML-based analyses of biological studies. Broad adoption of these recommendations help improve ML **assessment and reproducibility**.

Walsh, I., Fishman, D., Garcia-Gasulla, D. et al. DOME: recommendations for supervised machine learning validation in biology. Nat Methods 18, 1122–1127 (2021). https://doi.org/10.1038/s41592-021-01205-4

2. Gold Standard Datasets Task

AI-ready datasets are a needed for training and benchmarking ML methods. As such, main goal of this Task is to run a comprehensive effort around **gold standard datasets**, collecting both paper candidates with datasets on **selected domains** (e.g. genomics, transcriptomics, proteomics), as well as datasets that can be applied in ML in Life Sciences, specifically for **supervised learning** applications on **omics**.



3. Synthetic Data Task

Synthetic data represents **artificially generated information** that mimics real patient data while ensuring **privacy and confidentiality**.

In Life Sciences, the use of synthetic data facilitates **development opportunities** by enabling researchers to access vast and diverse datasets, thus accelerating drug discovery, disease modelling, and personalised medicine.

The Task Force on synthetic data is focused on developing **frameworks, guidelines and best practices** for **generating, evaluating, and using** synthetic data in Life Sciences.

Notable outputs:

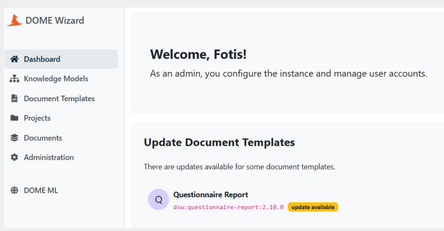
- establishment of a synthetic dataset **catalogue** aligned with the EDAM ontology,
- the definition of a **metadata model** for the publication of FAIR synthetic datasets,
- the inception of a synthetic data **registry**,
- the execution of a scoping review focused on synthetic data **evaluation metrics**,
- the implementation of a **community survey** to gauge the utilisation of synthetic data.

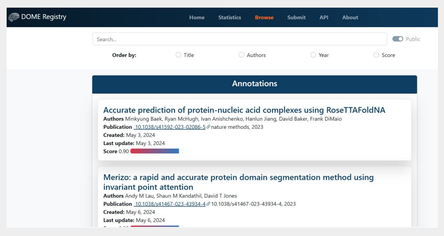
Queralt-Rosinach, Núria, et al. "Infrastructure for Synthetic Health Data." BioHackrXiv, 2023

DOME Strategic Implementation Study: A framework to standardize ML in Life Sciences

Three major activities:

1. Implement a **framework** around the DOME recommendations, including a registry to capture DOME-related information from existing and future literature, and a low-barrier software tool that will allow for the easy creation of DOME annotation by the wider community.
2. **Review** and **connect** the DOME recommendations in light of the particular needs, requirements and expectations of relevant ELIXIR communities, thus establishing a pathway towards **adoption**.
3. Create a clear **engagement plan** to **stakeholders** beyond ELIXIR, linking to relevant global efforts and initiatives, including the RDA ML interest groups, the NIH AI activities and efforts in Industry (such as the Pistoia Alliance).





Contact

Fotis Psomopoulos, *INAB|CERTH, GR*
Emidio Capriotti, *UNIBO, IT*
Núria Queralt Rosinach, *LUMC, NL*
Davide Cirillo, *BSC, ES*
Leyla Jael Castro, *ZBMed*
Silvio Tosatto, *UniPD, IT*
the ELIXIR ML Focus Group members

ml_focus_group@elixir-europe.org website: https://elixir-europe.org/focus-groups/machine-learning

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